



Era of experiential and heuristic learning

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Dear Editor:

As artificial intelligence advances from simulation and data-driven learning to workflow-integrated systems, formal education must undergo a parallel transformation. The changing workplace makes it increasingly apparent that traditional models, that is, front-loading content delivery through lectures, textbooks, and examinations, no longer suffice to prepare students for an uncertain, innovation-driven economy. Following a longstanding trend toward authentic modes of assessment, we propose that higher education embrace an *Era of Experiential and Heuristic Learning*, in which knowing and doing become inseparable—and in which students are taught how to think with AI, not just how to do with AI.

For decades, instruction relied on “simulated” environments (e.g., lecture halls, problem sets, and timed tests) that rewarded performance under controlled conditions rather than fluency with real-world ambiguity and consequence. Project-based initiatives, internships, and capstone courses began to bridge this gap, but high-quality opportunities have too often been confined to well-resourced institutions with strong industry ties. To democratize access, experiential activities must be woven into every semester and every cohort—not offered as optional add-ons, but as core curricular elements. Achieving this requires flexible partnerships with local companies, startups, and community

organizations. Local partners can contribute challenges, data sets, mentorship hours, or guest critiques, allowing a wider ecosystem to engage as co-designers and participants in students’ learning journeys.

For experiential learning to reach its full potential, it must engage student populations across a wide range of institutions and contexts. Faculty, practitioners, and organizational partners should co-design curricula that embed iterative feedback, structured reflection, and multi-semester continuity. At the same time, institutions can adopt modular collaboration models with local employers; enabling every student—regardless of the institution’s size or budget—to gain meaningful hands-on exposure to real-world workflows. To encourage adoption at scale, accrediting bodies and funding agencies should consider prioritizing proposals and programs that meaningfully integrate such experiential designs.

Parallel to curriculum redesign, assessment practices must evolve to reflect the reality of generative AI tools. Learning goals and evaluation criteria can often be organized around three types of knowledge: (i) factual knowledge, (ii) procedural knowledge, and (iii) metacognitive skills—the ability to plan, monitor, and reflect on one’s own thinking. Long-standing frameworks such as inquiry-based and project-based pedagogy remain highly relevant. Generative AI does not replace learning objectives focused on higher order thinking; rather, it redefines how students can leverage automation to achieve them. We recommend that programs begin explicitly incorporating “AI fluency” rubrics into their assessments, distinguishing between strategic AI use and passive reliance.

For example, we have presented comparisons of traditional assessment items, as they might appear in the pedagogically “good” or “bad” futures. In the “bad” future, an AI agent simply provides a (possibly hallucinated) solution to the assessment item. In the “good” future, students guide the AI tool through a “graph of thoughts” and verifications to arrive at a pedagogically meaningful solution (<https://wolfram.com/TransformingTeaching>). There is no question that the student treating the AI tool as a resource to which they can delegate implementation details and focus on strategic thinking

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is more pedagogically satisfying than copy-pasting traditional assessment items into an AI chat session. Indeed, a recent article emphasized the dangers of cognitive offloading without critical thinking (Gerlich 2025). There are suggestions to incorporate innovative ways to incorporate AI into traditional assessment tasks, such as prompt techniques. Review articles that provide a taxonomy of language model prompting techniques (Chen et al. 2025) can be used as inspiration for new innovative ways that students can use AI to emphasize strategic thinking on traditional assessment items. Designing effective prompts requires students to engage in advanced planning, much like pseudo-code development. In addition to benefiting the student, specifying intermediate steps also helps guide contemporary AI systems through complex solution spaces, leading to more accurate outcomes.

However, the point of the present note is to suggest that educators must move beyond traditional assessment items and go further than many have already. While it is worth considering that very specific learning goals do still align with a quick traditional assessment, authors in education have long emphasized the need for more authentic forms of assessment. The point we wish to emphasize is that neither educational institutions nor the employers for which we are preparing the future workforce are likely to have all the necessary expertise and resources to create these authentic learning experiences alone. While existing efforts to implement authentic assessment practices should be commended, we believe that the reality of generative AI capabilities reveals the need for a serious expansion of opportunities for the co-design of new learning experiences.

As a practical reality of the intersection of this new AI future and the past modes of assessment, consider the problem of access to AI technology. Without careful design of the actual assessment criteria and learning goals, we risk tying evaluations of student performance to what tools they can access. To address this risk, AI-usage disclosures, institution-provided access, and assessments that can differentiate the student's strategic input to the process from the final, tool-assisted output become critical.

Luckily, the last several decades of thought among education practitioners have already developed some conceptual tools for framing these new, especially pronounced realities. The need for (i) alignment of assessment and learning goals, (ii) robust professional development for practitioners, (iii) open-ended summative assessment experiences, and (iv) timely and iterative feedback are not new ideas. What is new is the necessity for these principles to be enforced not only at the course or department level but through coordinated institutional strategy and cross-sector partnerships.

To illustrate an *Era of Experiential and Heuristic Learning*, we offer concise vignettes across diverse domains. In a quantum materials workshop, students simulate finite 1D chains and 2D lattices of periodic wells, compute eigenenergies for

varying tunneling strengths to observe discrete levels splitting into minibands and continuous dispersion surfaces with van Hove singularities, then reflect and codify heuristics (“stronger coupling widens bands” and “larger lattices approximate continuum”) that cement both computational fluency and physical intuition. Or, as another example, an unmanned aerial-system capstone alternates roles between ground-crew diagnosticians using flowcharts and oscilloscopes to isolate faults and develop quick-fix rules (“if altimeter zig-zags but GPS is steady, swap the barometric sensor first”), and autonomous-mission pilots who adjust flight parameters in real time (“when wind gusts exceed 15 mph, slow descent rate by 10%”). Debriefs surface cross-cutting heuristics that link hardware reliability to mission success. Each scenario embeds iterative cycles of doing, feedback, reflection, and heuristic abstraction, hallmarks of experiential learning, and demonstrates how AI-enabled dashboards and simulation environments can partner in strategic problem solving rather than serve as repetitive answer tools.

By embedding sustained experiential learning throughout the curriculum and by rethinking assessment in light of generative AI's capabilities, we can cultivate practitioners who are both fluent with emerging technologies and adept at higher order problem solving. This dual transformation—of what we teach and how we evaluate—will better equip graduates to navigate and lead in a rapidly changing world. If educators and employers do not come together to address the challenges posed by new technologies collaboratively, we risk training students for an economic reality that no longer exists.

Curmudgeon Corner Curmudgeon Corner is a short opinionated column on trends in technology, arts, science and society, commenting on issues of concern to the research community and wider society. Whilst the drive for super-human intelligence promotes potential benefits to wider society, it also raises deep concerns of existential risk, thereby highlighting the need for an ongoing conversation between technology and society. At the core of Curmudgeon concern is the question: What is it to be human in the age of the AI machine? -Editor.

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Declarations

Conflict of interest The authors declare no competing interests.

References

Academic Innovation, Wolfram Research. Transforming teaching. See this notebook for more info. <https://wolfr.am/TransformingTeaching>

- Chen B et al (2025) Unleashing the potential of prompt engineering for large language models. *Patterns* 6(6):101260. <https://doi.org/10.1016/j.patter.2025.101260>
- Gerlich M (2025) AI tools in Society: impacts on cognitive offloading and the future of critical thinking. *Societies*. <https://doi.org/10.3390/soc15010006>

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